

Food Label Detective Agent

An agentic AI that reads food labels, explains them in plain English, and flags what matters to you.

Track: Agentic AI | **Consumer Safety & Nutrition** | **Open Food Facts**

Capstone Project — Agentic AI

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THE PROBLEM

Food labels are hard to read — and the stakes are personal

Cryptic ingredients. E-numbers and chemical names mean little to shoppers.

Hidden allergens. Milk, nuts and soy hide inside derivatives and 'traces'.

Manual diet checks. Vegan / low-sugar goals require reading every label.

Data overload. Nutrition tables need arithmetic to judge at a glance.

Real-world impact

A practical assistant for an everyday need.

Faster, safer choices at the shelf.

Human stays in control for sensitive calls.

Informs — never decides for you

DATASET & REFERENCE SOURCE

Live product data, with an offline safety net

Open Food Facts API

Primary source. Live ingredients, allergens, nutrition.

Rich fields. NOVA processing group + Nutri-Score.

Global. Lookup by barcode or product name.

Synthetic Starter CSV

10 products, same schema as the API.

Automatic fallback when offline or code missing.

Deterministic — makes the tests reproducible.

Clearly labelled as synthetic starter data · fields: ingredients, allergens, traces, sugar/salt/fat per 100g, NOVA, Nutri-Score

SYSTEM WORKFLOW

From a barcode to an actionable answer

1

Understand

Parse the query & user goal

2

Ask

Request missing info if needed

3

Tools + Memory

Open Food Facts + SQLite prefs

4

Analyse

Allergens, additives & risk score

5

Guardrails

Alerts, disclaimer, human-in-loop

6

Explain + Log

Plain English + JSONL trail

UI (Streamlit / CLI) → Agent Controller → Tools · Memory · LLM → Guardrail Check → Final Answer + Logs

AI / AGENT INNOVATION

A real agent — six cooperating modules



Clarifying questions

Asks for a product instead of failing.



Tool use

Calls Open Food Facts + CSV fallback.



Memory

Remembers diet & allergies in SQLite.



Reasoning

Transparent 0–100 risk score, itemised.



Guardrails

Allergy alerts, disclaimer, human-in-loop.



LLM explanation

Claude writes plain English (template fallback).

Why AI here? The LLM makes technical data understandable, while the rule-based engine keeps safety-critical logic auditable — AI where it adds value, not for decoration.

PROTOTYPE / DEMO

The result card — real output for a milk-allergic vegan user

🕵️ Food Label Detective · query: “Nutella”

Nutella Hazelnut Spread

🚫 BEST AVOIDED **Risk score 75 / 100 (high)**

⚠ **ALLERGY ALERT: may contain milk, which you asked to avoid.**

Not vegan-friendly — contains milk.

High in sugar, saturated fat & total fat.

Additive: palm oil. Ultra-processed (NOVA 4).

Recommendation: best avoided given your settings.

Runs anywhere

Streamlit UI + full CLI.

Zero-setup: works with no API key.

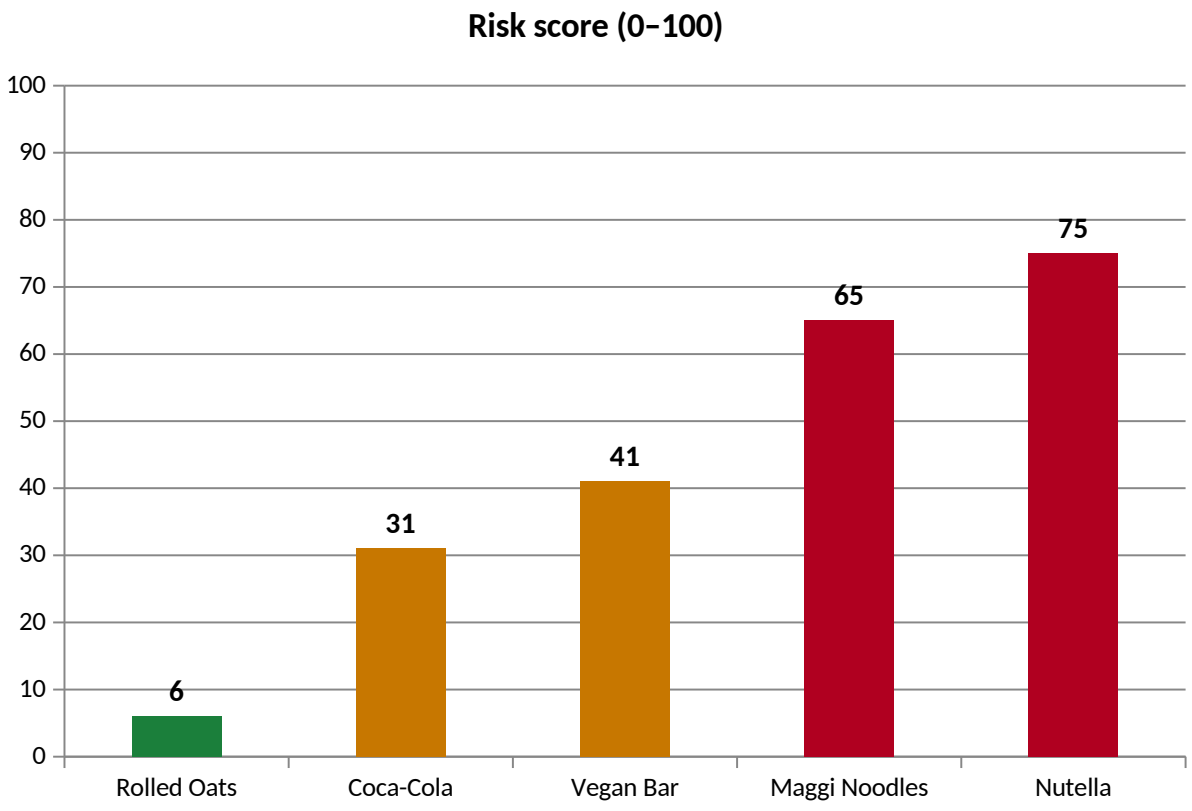
Set ANTHROPIC_API_KEY for Claude.

Every score shown, never a black box.

You make the final call

RESULTS & SAMPLE OUTPUTS

Correct risk bands across live & synthetic products



Product	Source	Advice
Nutella	synthetic	**avoid**
Rolled Oats	synthetic	ok
Coca-Cola	live API	moderate
Vegan Protein Bar	synthetic	**avoid**
Maggi Noodles	live API	**avoid**

10 / 10 scenario tests pass · runs verified offline & against live API

LIMITATIONS & RESPONSIBLE USE

Honest about what it can’t do

Crowd-sourced data. Some products have gaps — flagged, not guessed.

Keyword detection. Can miss unusual ingredient names.

General thresholds. Not personalised medical advice.

LLM dependence. Richest explanations need an API key.

Responsible use

This tool provides general information from food labels only. It is NOT medical or dietary advice.

People with allergies must confirm against the physical package. The agent never guarantees a product is safe — the final decision always stays with the human.

FUTURE IMPROVEMENTS

Where it goes next

Label OCR

Read a photo of a label for products not in the database.

Barcode scanning

Scan directly from the phone camera in the UI.

Personalised limits

e.g. diabetic-specific sugar thresholds.

Cited guidance (RAG)

Vector store of nutrition sources for richer answers.

Multi-language

Explanations in the user's language.

History & compare

Track scanned products and compare alternatives.

Trustworthy and understandable

A complete agentic workflow — input → tools & memory → guardrailed, explained output with logs. An auditable rule-based safety engine paired with an LLM explainer, keeping humans in control.

10/10 tests pass

Live + offline verified

Responsible by design

Thank you · Vatsal Jain · Vidit Jain · Vanshika Nehra